

Brutal Practice Paper B06 - Soil / Rational Numbers / Winds & Cyclones / Symmetry

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A 250 mL sample of water drains through a sandy-soil column in 10 minutes, while the same volume needs 50 minutes through a clayey column. By how much (in mL/min) is the sandy soil's percolation rate greater than the clayey soil's? [\[Soil\]](#)
 - (A) 20 mL/min
 - (B) 40 mL/min
 - (C) 30 mL/min
 - (D) 25 mL/min
 - (E) 5 mL/min
 - (F) 35 mL/min

2. Find the value of $(-3/4) + (5/6)$. [\[Rational Numbers\]](#)
 - (A) $1/2$
 - (B) $-1/12$
 - (C) $1/12$
 - (D) $-19/12$
 - (E) $2/10$
 - (F) $8/10$

3. A cyclone is moving toward the coast at 15 km/h and is now 240 km away. If it keeps this speed, in how many hours will it reach the coast? [\[Winds, Storms & Cyclones\]](#)
 - (A) 16 hours
 - (B) 255 hours
 - (C) 3600 hours
 - (D) 8 hours
 - (E) 15 hours
 - (F) 12 hours
 - (G) 24 hours

4. A figure has rotational symmetry of order 8. What is the smallest angle through which it can be turned to look exactly the same? [\[Symmetry\]](#)
 - (A) 90 deg
 - (B) 40 deg
 - (C) 60 deg
 - (D) 8 deg
 - (E) 360 deg
 - (F) 45 deg
 - (G) 72 deg

5. In a percolation test, 320 mL of water passes through a soil in 16 minutes. A second soil lets water through at exactly half this rate. How long will 320 mL take in the second soil? [\[Soil\]](#)

- (A) 32 minutes
- (B) 16 minutes
- (C) 8 minutes
- (D) 20 minutes
- (E) 48 minutes
- (F) 64 minutes

6. Subtract: $(-2/3) - (-7/9)$. [\[Rational Numbers\]](#)

- (A) $-1/9$
- (B) $13/9$
- (C) $1/9$
- (D) $5/6$
- (E) $1/3$
- (F) $-5/9$
- (G) $-13/9$

7. A storm's wind blows at 54 km/h. Using 1 km = 1000 m and 1 hour = 3600 s, what is the speed in m/s?

[\[Winds, Storms & Cyclones\]](#)

- (A) 0.015 m/s
- (B) 90 m/s
- (C) 15 m/s
- (D) 150 m/s
- (E) 194.4 m/s
- (F) 54 m/s
- (G) 18 m/s

8. A windmill looks exactly the same after every 30-deg rotation. What is the order of its rotational symmetry? [\[Symmetry\]](#)

- (A) 330
- (B) 12
- (C) 30
- (D) 36
- (E) 6
- (F) 11
- (G) 3

9. A soil pit is 1.2 m deep. The topsoil (A-horizon) is 25 cm, the subsoil (B-horizon) is three times the topsoil, and the rest is the C-horizon. How thick is the C-horizon? [\[Soil\]](#)

- (A) 45 cm
- (B) 70 cm
- (C) 15 cm
- (D) 95 cm
- (E) 75 cm
- (F) 20 cm
- (G) 100 cm

10. Multiply $(-5/8)$ by $(12/-15)$ and give the answer in lowest terms. [\[Rational Numbers\]](#)

- (A) 2
- (B) $3/4$
- (C) -2
- (D) $5/8$
- (E) $1/2$
- (F) $60/120$
- (G) $-1/2$

11. During the daytime the land heats up faster than the sea. In which direction does the breeze blow, and why? [\[Winds, Storms & Cyclones\]](#)

- (A) From land to sea, because the sea is warmer by day
- (B) Straight upward, because hot air only rises
- (C) There is no breeze during the day
- (D) From land to sea, because warm air over the sea rises
- (E) From sea to land, because the sea is hotter than the land
- (F) From sea to land, because warm air over land rises and cooler sea air moves in

12. How many lines of symmetry does a regular hexagon (6 equal sides) have? [\[Symmetry\]](#)

- (A) 8
- (B) 6
- (C) 2
- (D) 1
- (E) 12
- (F) 3
- (G) 0

13. A clayey soil sample weighs 250 g when wet and 200 g after drying in the sun. What percent of the WET sample's mass was water? [\[Soil\]](#)

- (A) 50%
- (B) 12.5%
- (C) 40%
- (D) 20%
- (E) 80%
- (F) 5%
- (G) 25%

14. Divide $(-9/16)$ by $(3/-8)$. [\[Rational Numbers\]](#)

- (A) $3/2$
- (B) $-2/3$
- (C) $27/128$
- (D) $2/3$
- (E) $-3/2$
- (F) $3/8$
- (G) $-27/128$

15. A barometer reads 1000 units in calm weather and 960 units as a cyclone approaches. By what percentage did the air pressure drop? [\[Winds, Storms & Cyclones\]](#)

- (A) 40%
- (B) 0.4%
- (C) 4%
- (D) 96%
- (E) 2%
- (F) 10%
- (G) 6%

16. A regular pentagon is rotated by its smallest symmetry angle again and again. Does it ever land EXACTLY on a half turn (180 deg)? [\[Symmetry\]](#)

- (A) Yes, after 4 rotations
- (B) Yes, after exactly 3 rotations
- (C) Yes, after exactly 2 rotations
- (D) Yes, after 1 rotation
- (E) Yes, after exactly 5 rotations
- (F) No, it never lands exactly on 180 deg
- (G) Yes, after 2.5 rotations

17. Loam is a mixture of sand, silt and clay. A 400 g loam sample contains 180 g sand and 120 g silt, the rest being clay. What fraction of the sample is clay? [\[Soil\]](#)

- (A) $\frac{3}{10}$
- (B) $\frac{1}{4}$
- (C) $\frac{7}{20}$
- (D) $\frac{9}{20}$
- (E) $\frac{2}{5}$
- (F) $\frac{1}{3}$
- (G) $\frac{1}{5}$

18. Write the rational number $\frac{18}{-24}$ in standard form (positive denominator, lowest terms). [\[Rational Numbers\]](#)

- (A) $-\frac{3}{4}$
- (B) $\frac{9}{12}$
- (C) $-\frac{9}{12}$
- (D) $\frac{3}{4}$
- (E) $-\frac{6}{8}$
- (F) $\frac{4}{3}$

19. What is the MAIN cause of winds on Earth? [\[Winds, Storms & Cyclones\]](#)

- (A) Waves moving on the sea
- (B) Birds flapping their wings
- (C) Trees swaying back and forth
- (D) Uneven heating of the Earth's surface
- (E) Clouds bumping together
- (F) The spinning of electric fans

20. Among the capital letters A, B, F, H, N, O, how many have AT LEAST one line of symmetry? [\[Symmetry\]](#)

- (A) 0
- (B) 6
- (C) 4
- (D) 3
- (E) 5
- (F) 2
- (G) 1

21. Nature forms about 1 cm of topsoil in roughly 200 years. About how many years are needed to build a 15 cm layer of topsoil? [\[Soil\]](#)

- (A) 1500 years
- (B) 2000 years
- (C) 3000 years
- (D) 30000 years
- (E) 13 years
- (F) 215 years
- (G) 200 years

22. Which is greater: $-\frac{3}{5}$ or $-\frac{2}{3}$? [\[Rational Numbers\]](#)

- (A) $-\frac{2}{3}$
- (B) they are equal
- (C) $-\frac{3}{5}$
- (D) $-\frac{4}{6}$
- (E) $-\frac{6}{10}$
- (F) cannot be compared
- (G) $-\frac{1}{2}$

23. At the very centre of a cyclone, called the eye, what is the weather like? [\[Winds, Storms & Cyclones\]](#)

- (A) Total darkness with no air at all
- (B) Freezing snow and hail
- (C) The heaviest rain of the whole storm
- (D) The strongest winds of the whole storm
- (E) Calm, with light winds and a fairly clear sky
- (F) Constant lightning only
- (G) The coldest part of the storm

24. Which capital letter has rotational symmetry of order 2 but NO line of symmetry? [\[Symmetry\]](#)

- (A) M
- (B) T
- (C) X
- (D) S
- (E) O
- (F) H
- (G) A

- 25.** A field test pours 1.5 litres of water into a soil ring and it drains in 5 minutes. Taking 1 litre = 1000 mL, what is the percolation rate in mL/min? [\[Soil\]](#)
- (A) 0.3 mL/min
 - (B) 300 mL/min
 - (C) 3 mL/min
 - (D) 150 mL/min
 - (E) 1500 mL/min
 - (F) 7500 mL/min
 - (G) 30 mL/min
- 26.** Find the rational number lying exactly halfway between $\frac{1}{3}$ and $\frac{1}{2}$. [\[Rational Numbers\]](#)
- (A) $\frac{2}{5}$
 - (B) $\frac{5}{6}$
 - (C) $\frac{1}{12}$
 - (D) $\frac{5}{12}$
 - (E) $\frac{2}{12}$
 - (F) $\frac{3}{10}$
- 27.** A weather balloon drifts with the wind and covers 36 km in 1.5 hours. What is the wind speed in km/h? [\[Winds, Storms & Cyclones\]](#)
- (A) 18 km/h
 - (B) 48 km/h
 - (C) 54 km/h
 - (D) 30 km/h
 - (E) 12 km/h
 - (F) 24 km/h
 - (G) 37.5 km/h
- 28.** The 12 hour-marks of a clock face are spaced evenly. Treating only these marks, what is the order of rotational symmetry and the smallest rotation angle? [\[Symmetry\]](#)
- (A) Order 12, angle 12 deg
 - (B) Order 12, angle 360 deg
 - (C) Order 60, angle 6 deg
 - (D) Order 24, angle 15 deg
 - (E) Order 30, angle 12 deg
 - (F) Order 12, angle 30 deg
- 29.** Soil X drains 240 mL in 6 minutes; soil Y drains 240 mL in 8 minutes. Which soil is more likely SANDY, and what is its percolation rate? [\[Soil\]](#)
- (A) Soil Y, 40 mL/min
 - (B) Soil Y, 30 mL/min
 - (C) Soil Y, 8 mL/min
 - (D) Soil X, 80 mL/min
 - (E) Soil X, 6 mL/min
 - (F) Soil X, 40 mL/min
 - (G) Soil X, 30 mL/min

- 30.** The additive inverse of a rational number is $7/-9$. What is the number, written in standard form? [\[Rational Numbers\]](#)
- (A) $-1/9$
 - (B) $1/9$
 - (C) $7/9$
 - (D) $9/7$
 - (E) $7/-9$
 - (F) $-9/7$
 - (G) $-7/9$
- 31.** When you blow hard between two hanging sheets of paper, they swing TOWARDS each other. Why? [\[Winds, Storms & Cyclones\]](#)
- (A) Fast-moving air between them has lower pressure, so the higher outside pressure pushes them together
 - (B) The blown air pushes them apart from inside
 - (C) Gravity bends them inward
 - (D) The paper sheets attract each other like magnets
 - (E) Your breath is heavier than the paper
 - (F) Static electricity pulls them together
 - (G) The air between them becomes solid
- 32.** A square has how many lines of symmetry, and what is its order of rotational symmetry? [\[Symmetry\]](#)
- (A) 1 line, order 4
 - (B) 2 lines, order 2
 - (C) 2 lines, order 4
 - (D) 4 lines, order 2
 - (E) 4 lines, order 4
 - (F) 8 lines, order 4
- 33.** A pot holds 600 g of moist soil containing 15% water by mass. After a hot day it loses 54 g of water by evaporation. How many grams of water remain in the soil? [\[Soil\]](#)
- (A) 90 g
 - (B) 36 g
 - (C) 546 g
 - (D) 9 g
 - (E) 30 g
 - (F) 81 g
- 34.** What is the reciprocal of the product $(-4/5) \times (10/-3)$? [\[Rational Numbers\]](#)
- (A) $15/40$
 - (B) $-8/3$
 - (C) $8/3$
 - (D) $3/8$
 - (E) $-3/8$
 - (F) $40/15$
 - (G) $5/4$

35. The summer monsoon that brings most of India's rain blows mainly from which direction? [\[Winds, Storms & Cyclones\]](#)

- (A) From the north-east, off the land toward the sea
- (B) From the east, off dry desert land
- (C) From the south pole
- (D) From the north pole toward the equator only
- (E) From the west, but only over mountains
- (F) Straight down from directly overhead
- (G) From the south-west, off the sea toward the land

36. A figure has rotational symmetry of order 6. Through what TOTAL angle has it turned after 4 of its smallest rotations? [\[Symmetry\]](#)

- (A) 24 deg
- (B) 120 deg
- (C) 240 deg
- (D) 360 deg
- (E) 90 deg
- (F) 300 deg
- (G) 60 deg

37. Paddy (rice) is grown in fields kept flooded with standing water. Which soil type best suits paddy because it holds water well and drains slowly? [\[Soil\]](#)

- (A) Sandy soil
- (B) Gravelly soil
- (C) Clayey soil
- (D) Coarse sand
- (E) Silty sand
- (F) Loamy soil
- (G) Stony soil

38. Evaluate $(\frac{2}{3}) + (-\frac{5}{6}) - (\frac{1}{4})$. [\[Rational Numbers\]](#)

- (A) $-\frac{5}{12}$
- (B) $-\frac{1}{12}$
- (C) $-\frac{11}{12}$
- (D) $\frac{5}{12}$
- (E) $-\frac{1}{4}$
- (F) $\frac{1}{4}$
- (G) $\frac{11}{12}$

39. A cyclone travels at 20 km/h for 3 hours, then slows to 12 km/h for the next 5 hours. What total distance does it cover? [\[Winds, Storms & Cyclones\]](#)

- (A) 96 km
- (B) 128 km
- (C) 60 km
- (D) 160 km
- (E) 256 km
- (F) 32 km
- (G) 120 km

40. An equilateral triangle has order-3 rotational symmetry. What is its smallest angle of rotation, and how many lines of symmetry does it have? [\[Symmetry\]](#)
- (A) 90 deg, 3 lines
 - (B) 120 deg, 1 line
 - (C) 120 deg, 6 lines
 - (D) 60 deg, 1 line
 - (E) 120 deg, 3 lines
 - (F) 60 deg, 3 lines
 - (G) 180 deg, 3 lines
41. Three trials give percolation rates of 28, 32 and 30 mL/min for a loamy soil. A clayey soil averages 6 mL/min. The loamy soil's average rate is how many times the clayey soil's? [\[Soil\]](#)
- (A) 4 times
 - (B) 6 times
 - (C) 90 times
 - (D) 10 times
 - (E) 5 times
 - (F) 24 times
42. A diver descends $\frac{3}{4}$ of a metre each second. Taking below the surface as negative, what rational number gives the diver's position after 6 seconds? [\[Rational Numbers\]](#)
- (A) $\frac{9}{2}$ m
 - (B) $-\frac{9}{2}$ m
 - (C) $-\frac{3}{2}$ m
 - (D) $-\frac{27}{4}$ m
 - (E) $-\frac{2}{3}$ m
 - (F) $-\frac{9}{4}$ m
 - (G) $\frac{9}{4}$ m
43. Why does warm air rise above cooler air? [\[Winds, Storms & Cyclones\]](#)
- (A) Cool air is always pulled upward by the Sun
 - (B) Warm air has no weight at all
 - (C) Cool air pushes warm air down to the ground
 - (D) Warm air is heavier and sinks under cool air
 - (E) Warm air is attracted to the clouds
 - (F) Warm air expands, becomes less dense (lighter), and floats above the denser cool air
44. A point lies 5 cm to the right of a vertical mirror line. Where is its mirror image? [\[Symmetry\]](#)
- (A) 5 cm to the left of the line
 - (B) 2.5 cm to the left of the line
 - (C) 5 cm to the right of the line
 - (D) exactly on the line
 - (E) 10 cm to the right of the line
 - (F) 5 cm above the line

45. A 5 kg soil sample is found to contain 250 g of plastic waste. What percentage of the sample, by mass, is plastic? [\[Soil\]](#)
- (A) 25%
 - (B) 2%
 - (C) 50%
 - (D) 5%
 - (E) 20%
 - (F) 0.5%
46. Arrange in ascending order (smallest first): $-1/2$, $3/4$, $-3/8$, 0. [\[Rational Numbers\]](#)
- (A) 0, $-3/8$, $-1/2$, $3/4$
 - (B) $-1/2$, 0, $-3/8$, $3/4$
 - (C) $-3/8$, $-1/2$, $3/4$, 0
 - (D) $-3/8$, $-1/2$, 0, $3/4$
 - (E) $-1/2$, $-3/8$, $3/4$, 0
 - (F) $-1/2$, $-3/8$, 0, $3/4$
 - (G) $3/4$, 0, $-3/8$, $-1/2$
47. A cyclone's winds reach 50 m/s. What is this speed in km/h? (1 m/s = 3.6 km/h) [\[Winds, Storms & Cyclones\]](#)
- (A) 50 km/h
 - (B) 1800 km/h
 - (C) 46.4 km/h
 - (D) 180 km/h
 - (E) 53.6 km/h
 - (F) 13.9 km/h
 - (G) 150 km/h
48. Can a figure have rotational symmetry whose smallest rotation angle is exactly 50 deg? (The angle must divide 360 deg exactly.) [\[Symmetry\]](#)
- (A) Yes, its order would be 50
 - (B) No, because 360 is not exactly divisible by 50
 - (C) Yes, its order would be 5
 - (D) Yes, its order would be 360
 - (E) Yes, its order would be 7
 - (F) No, because 50 is an even number
 - (G) Yes, its order would be 7.2
49. Water percolates downward through a sandy soil at 4 cm per minute. How deep will the wetting front be after 3.5 minutes? [\[Soil\]](#)
- (A) 7.5 cm
 - (B) 12 cm
 - (C) 11.5 cm
 - (D) 14 cm
 - (E) 1.14 cm
 - (F) 16 cm
 - (G) 0.875 cm

50. Find $(-2/3) \times (9/-4) \times (1/3)$. [\[Rational Numbers\]](#)

- (A) $-1/2$
- (B) $1/6$
- (C) $2/3$
- (D) $-1/6$
- (E) $-3/2$
- (F) $1/2$

51. A tightly closed empty plastic bottle is carried from a high hill down to the plains, where it gets slightly crushed. The best explanation is that... [\[Winds, Storms & Cyclones\]](#)

- (A) The plains are colder, freezing the bottle
- (B) Gravity is stronger in the plains and squashes it
- (C) Air pressure is higher in the plains, pressing the bottle inward
- (D) The bottle absorbs water from the air
- (E) Plastic shrinks in sunlight on the plains
- (F) There is no air in the plains
- (G) Air pressure is lower in the plains, so the bottle bursts

52. A perfectly round wheel is folded onto itself along a straight crease through its centre. Counting every such possible crease, how many lines of symmetry does the circle have? [\[Symmetry\]](#)

- (A) 2
- (B) Infinitely many
- (C) 0
- (D) 8
- (E) 1
- (F) 360

53. Which soil horizon is the darkest, richest in humus, and supports most of the plant roots? [\[Soil\]](#)

- (A) C-horizon
- (B) Parent rock layer
- (C) The water table
- (D) A-horizon (topsoil)
- (E) Bedrock
- (F) B-horizon (subsoil)
- (G) D-horizon

54. How many rational numbers lie between $1/4$ and $1/2$? [\[Rational Numbers\]](#)

- (A) 1
- (B) 2
- (C) 3
- (D) four
- (E) infinitely many
- (F) none
- (G) exactly 7

55. A cyclone is 350 km away and moving toward a town at 25 km/h. A safe evacuation needs 8 hours. Is there enough time, and by how much? [\[Winds, Storms & Cyclones\]](#)
- (A) Yes, with 14 hours to spare
 - (B) No, short by 6 hours
 - (C) Yes, with 6 hours to spare
 - (D) Yes, with 2 hours to spare
 - (E) No, short by 2 hours
 - (F) No, short by 14 hours
56. A rectangle that is NOT a square has how many lines of symmetry, and what order of rotational symmetry? [\[Symmetry\]](#)
- (A) 1 line, order 2
 - (B) 4 lines, order 4
 - (C) 2 lines, order 4
 - (D) 2 lines, order 1
 - (E) 4 lines, order 2
 - (F) 2 lines, order 2
 - (G) 0 lines, order 2
57. In a soil mix the ratio of sand to clay is 3 : 2 and the total mass is 750 g. How many grams of clay are present? [\[Soil\]](#)
- (A) 375 g
 - (B) 450 g
 - (C) 500 g
 - (D) 300 g
 - (E) 250 g
 - (F) 200 g
 - (G) 150 g
58. If $-5/7 = x/49$, what is the value of x? [\[Rational Numbers\]](#)
- (A) 35
 - (B) -7
 - (C) -49
 - (D) -245
 - (E) -5
 - (F) 7
 - (G) -35
59. During a thunderstorm, which of these is the SAFEST action? [\[Winds, Storms & Cyclones\]](#)
- (A) Shelter under a tall, lone tree
 - (B) Stay indoors, away from windows and metal objects
 - (C) Stand in an open field to watch
 - (D) Fly a kite to test the wind
 - (E) Hold up a metal-framed umbrella
 - (F) Touch metal pipes to feel safe

60. A regular polygon has a smallest rotation angle of 24 deg. How many sides does it have? [\[Symmetry\]](#)

- (A) 24
- (B) 15 deg
- (C) 12
- (D) 15
- (E) 8
- (F) 336
- (G) 10

61. Soil takes in heat by day and releases it by night. If the soil surface warms from 22 deg C to 40 deg C between 6 a.m. and noon, what is the average rate of temperature rise per hour? [\[Soil\]](#)

- (A) 3 deg C per hour
- (B) 2 deg C per hour
- (C) 3.5 deg C per hour
- (D) 40 deg C per hour
- (E) 18 deg C per hour
- (F) 6 deg C per hour

62. At night the temperature is $-5/2$ deg C; by noon it rises to $7/4$ deg C. By how many degrees did it rise?

[\[Rational Numbers\]](#)

- (A) $-3/4$ deg C
- (B) $17/4$ deg C
- (C) $12/4$ deg C
- (D) $-17/4$ deg C
- (E) $13/4$ deg C
- (F) $3/4$ deg C

63. You see a flash of lightning and hear the thunder 6 seconds later. If sound travels about 340 m/s, roughly how far away was the lightning? (1 km = 1000 m) [\[Winds, Storms & Cyclones\]](#)

- (A) about 57 km
- (B) about 0.057 km
- (C) about 20 km
- (D) about 2 km
- (E) about 0.34 km
- (F) about 2040 km
- (G) about 6 km

64. A figure looks the same ONLY after a complete 360-deg turn. What is its order of rotational symmetry?

[\[Symmetry\]](#)

- (A) Order 1
- (B) Infinitely many
- (C) Order 36
- (D) Order 0
- (E) Order 4
- (F) Order 360
- (G) Order 2

65. Equal 100 g dry samples of clay and sand are soaked with water. The clay holds 45 g of water and the sand holds 18 g. Per 100 g of soil, how much MORE water does the clay hold than the sand? [\[Soil\]](#)
- (A) 30 g
 - (B) 27 g
 - (C) 2.5 g
 - (D) 63 g
 - (E) 45 g
 - (F) 18 g
66. Which rational number is its own reciprocal AND is negative? [\[Rational Numbers\]](#)
- (A) 1
 - (B) 0
 - (C) 2
 - (D) $\frac{1}{2}$
 - (E) -2
 - (F) -1
 - (G) $-\frac{1}{2}$
67. A cyclone dumps 18 cm of rain on a flat field of area 200 square metres. Taking 1 cm of rain on 1 square metre as 10 litres, how many litres fall on the field? [\[Winds, Storms & Cyclones\]](#)
- (A) 3618 litres
 - (B) 218 litres
 - (C) 1800 litres
 - (D) 20000 litres
 - (E) 360 litres
 - (F) 36000 litres
 - (G) 3600 litres
68. Which capital letter has BOTH a vertical and a horizontal line of symmetry: A, B, X, M? [\[Symmetry\]](#)
- (A) none of them
 - (B) B and M
 - (C) B
 - (D) A
 - (E) M
 - (F) X
69. A soak pit must drain 600 mL of water in under 5 minutes. A soil percolates at 90 mL/min. Will it drain in time, and by how much (mL/min) does it fall short of or beat the needed rate? [\[Soil\]](#)
- (A) Yes, 30 mL/min to spare
 - (B) No, 510 mL/min too slow
 - (C) No, 120 mL/min too slow
 - (D) Yes, exactly on time
 - (E) No, 30 mL/min too slow
 - (F) Yes, 60 mL/min to spare

70. Simplify: $(-7/10) \div (14/-25)$. [\[Rational Numbers\]](#)

- (A) $-4/5$
- (B) $25/14$
- (C) $5/4$
- (D) $-25/14$
- (E) $175/140$
- (F) $-5/4$
- (G) $4/5$

71. At night the land cools faster than the sea. In which direction does the breeze then blow? [\[Winds, Storms & Cyclones\]](#)

- (A) From land to sea (a land breeze)
- (B) From sea to land (a sea breeze)
- (C) It stops blowing completely at night
- (D) Straight up into the clouds
- (E) From the sea straight down
- (F) From the sky down to the ground

72. A shape has order-5 rotational symmetry. Its smallest rotation is what fraction of a full turn? [\[Symmetry\]](#)

- (A) $1/360$
- (B) 72
- (C) $1/5$
- (D) $2/5$
- (E) $1/5$ deg
- (F) 5
- (G) $1/72$

73. Moist soil weighs 480 g; after fully drying it weighs 384 g. What was the moisture percentage of the original MOIST soil? [\[Soil\]](#)

- (A) 25%
- (B) 5%
- (C) 12.5%
- (D) 96%
- (E) 20%
- (F) 16%
- (G) 80%

74. Starting from $2/3$ on the number line, you move 5 whole units to the LEFT. What number do you reach?

[\[Rational Numbers\]](#)

- (A) $-13/3$
- (B) $17/3$
- (C) $13/5$
- (D) $-17/3$
- (E) $13/3$
- (F) $-7/3$

75. Hourly wind speeds were recorded as 12, 18, 21, 15 and 24 km/h. What is the average wind speed?

[Winds, Storms & Cyclones]

- (A) 24 km/h
- (B) 16 km/h
- (C) 20 km/h
- (D) 21 km/h
- (E) 15 km/h
- (F) 90 km/h
- (G) 18 km/h

76. A rangoli pattern looks identical after every 40-deg rotation. What is the order of its rotational symmetry? [Symmetry]

- (A) 36
- (B) 320
- (C) 8
- (D) 40
- (E) 9 deg
- (F) 9
- (G) 10

77. Which soil, having a balanced mixture of sand, silt and clay with moderate percolation, is described as the BEST for growing most crops? [Soil]

- (A) Clayey soil
- (B) Pure silt
- (C) Coarse sand
- (D) Loamy soil
- (E) Gravelly soil
- (F) Stony soil

78. Asha spends $\frac{2}{5}$ of her money on books and $\frac{1}{3}$ of the REMAINDER on snacks. What fraction of her original money is left? [Rational Numbers]

- (A) $\frac{3}{5}$
- (B) $\frac{2}{15}$
- (C) $\frac{1}{5}$
- (D) $\frac{2}{5}$
- (E) $\frac{7}{15}$
- (F) $\frac{4}{15}$

79. A little water is boiled in a metal can, the can is then sealed and cooled, and it collapses inward. Why?

[Winds, Storms & Cyclones]

- (A) The metal melts and folds over
- (B) Steam inside condenses, lowering the inside pressure, so outside air pressure crushes it
- (C) Cooling makes the can heavier
- (D) The can is pulled inward by gravity
- (E) The water inside expands and pushes the walls in
- (F) Heat makes metal shrink to nothing
- (G) Air rushes in and inflates it

- 80.** A square sheet of paper is folded so the two halves match exactly. Along how many DIFFERENT fold lines is this possible? [\[Symmetry\]](#)
- (A) 6
 - (B) 0
 - (C) 8
 - (D) 1
 - (E) 3
 - (F) 4
 - (G) 2
- 81.** In an 80 cm soil profile, the topsoil is $\frac{1}{8}$ of the depth and the subsoil is $\frac{3}{8}$ of the depth. How many cm is left for the deeper layers? [\[Soil\]](#)
- (A) 10 cm
 - (B) 40 cm
 - (C) 35 cm
 - (D) 50 cm
 - (E) 30 cm
 - (F) 60 cm
- 82.** By what rational number must $-\frac{8}{15}$ be multiplied to give $\frac{2}{5}$? [\[Rational Numbers\]](#)
- (A) $\frac{4}{3}$
 - (B) $-\frac{3}{4}$
 - (C) $-\frac{16}{75}$
 - (D) $-\frac{3}{5}$
 - (E) $\frac{3}{4}$
 - (F) $-\frac{4}{3}$
 - (G) $\frac{16}{75}$
- 83.** A gentle breeze blows at 12 km/h while a cyclone's wind blows at 144 km/h. The cyclone wind is how many TIMES the breeze? [\[Winds, Storms & Cyclones\]](#)
- (A) 11 times
 - (B) 12 times
 - (C) 132 times
 - (D) 12 km/h
 - (E) 156 times
 - (F) 6 times
 - (G) 1728 times
- 84.** A figure has order-9 rotational symmetry. After it has turned through 150 deg, by how many MORE degrees (in the same direction) must it turn to next look the same? [\[Symmetry\]](#)
- (A) 30 deg
 - (B) 40 deg
 - (C) 20 deg
 - (D) 10 deg
 - (E) 50 deg
 - (F) 9 deg
 - (G) 200 deg

85. A clayey soil percolates at 8 mL/min. How long, in minutes, will it take to drain 200 mL of water? [\[Soil\]](#)

- (A) 12.5 minutes
- (B) 25 minutes
- (C) 1600 minutes
- (D) 16 minutes
- (E) 8 minutes
- (F) 40 minutes

86. Which of these is the SMALLEST: $-4/5$, $-5/6$, $-3/4$? [\[Rational Numbers\]](#)

- (A) they are all equal
- (B) $-3/4$
- (C) $-1/2$
- (D) $-4/5$
- (E) 0
- (F) $-2/3$
- (G) $-5/6$

87. In a storm, strong wind blowing over a flat roof can lift it right off the house. This is because the fast wind above the roof creates... [\[Winds, Storms & Cyclones\]](#)

- (A) Higher pressure above, which pushes the roof down hard
- (B) Extra weight on top of the roof
- (C) Heat that melts the roof joints
- (D) Rain that floats the roof away
- (E) A magnetic pull on the roof tiles
- (F) Lower pressure above, so the higher pressure inside pushes the roof up

88. Which of these has rotational symmetry but NO line of symmetry: a scalene triangle, the letter S, a square, or an equilateral triangle? [\[Symmetry\]](#)

- (A) the letter S
- (B) a scalene triangle
- (C) the square and the S
- (D) a square
- (E) none of them
- (F) an equilateral triangle
- (G) all of them

89. A bare slope loses topsoil at 3 mm per year, while a grass-covered slope loses it at 0.5 mm per year. Over 20 years, how much MORE topsoil (in mm) does the bare slope lose? [\[Soil\]](#)

- (A) 10 mm
- (B) 60 mm
- (C) 50 mm
- (D) 70 mm
- (E) 35 mm
- (F) 25 mm
- (G) 2.5 mm

90. Evaluate $(-1/2) + (3/4) + (-5/8)$ and give the answer in lowest terms. [\[Rational Numbers\]](#)

- (A) $-3/4$
- (B) $1/8$
- (C) $-1/8$
- (D) $7/8$
- (E) $-3/8$
- (F) $-7/8$

91. When a cyclone warning is issued, which of these is NOT a sensible thing to do? [\[Winds, Storms & Cyclones\]](#)

- (A) Move animals to higher ground
- (B) Charge phones and torches
- (C) Store extra drinking water
- (D) Keep an emergency kit ready
- (E) Move to a safe cyclone shelter
- (F) Listen to official radio updates
- (G) Go to the beach to watch the high waves

92. Among the capital letters H, I, N, O, S and T, how many have rotational symmetry of order 2? [\[Symmetry\]](#)

- (A) 3
- (B) 0
- (C) 2
- (D) 1
- (E) 4
- (F) 5
- (G) 6

93. A dry clay block of mass 300 g can soak up water equal to 40% of its dry mass. What will the block weigh when fully saturated with water? [\[Soil\]](#)

- (A) 420 g
- (B) 460 g
- (C) 180 g
- (D) 120 g
- (E) 440 g
- (F) 300 g
- (G) 340 g

94. What is the distance on the number line between $-7/6$ and $5/3$? [\[Rational Numbers\]](#)

- (A) $5/6$
- (B) $3/2$
- (C) $17/3$
- (D) $17/6$
- (E) $1/2$
- (F) $3/6$

- 95.** Before a thunderstorm the temperature was 38 deg C; it then fell by 2 deg C every 10 minutes for 40 minutes as cool air rushed in. What was the temperature after the 40 minutes? [\[Winds, Storms & Cyclones\]](#)
- (A) 6 deg C
 - (B) 36 deg C
 - (C) 22 deg C
 - (D) 34 deg C
 - (E) 28 deg C
 - (F) 30 deg C
 - (G) 32 deg C
- 96.** A regular octagon (8 equal sides) has how many lines of symmetry, and what is its smallest angle of rotational symmetry? [\[Symmetry\]](#)
- (A) 8 lines, 90 deg
 - (B) 8 lines, 60 deg
 - (C) 16 lines, 45 deg
 - (D) 4 lines, 90 deg
 - (E) 8 lines, 45 deg
 - (F) 4 lines, 45 deg
- 97.** Four soils each drain 180 mL of water, taking 4, 6, 9 and 12 minutes. Which soil has the LOWEST percolation rate, and what is that rate? [\[Soil\]](#)
- (A) the 12-minute soil, 12 mL/min
 - (B) the 4-minute soil, 15 mL/min
 - (C) the 9-minute soil, 20 mL/min
 - (D) the 12-minute soil, 180 mL/min
 - (E) the 12-minute soil, 15 mL/min
 - (F) the 6-minute soil, 30 mL/min
- 98.** Express $-\frac{42}{-56}$ in standard form (lowest terms, positive denominator). [\[Rational Numbers\]](#)
- (A) $\frac{3}{4}$
 - (B) $\frac{6}{8}$
 - (C) $-\frac{6}{8}$
 - (D) $-\frac{3}{4}$
 - (E) $\frac{3}{-4}$
 - (F) $\frac{4}{3}$
 - (G) $\frac{7}{8}$
- 99.** Storm A blows at 90 km/h and storm B at 20 m/s. Using $1 \text{ m/s} = 3.6 \text{ km/h}$, which storm has the faster wind? [\[Winds, Storms & Cyclones\]](#)
- (A) Storm B, at 56 km/h
 - (B) Cannot be compared
 - (C) Storm A, at 90 km/h
 - (D) Storm B, at 20 km/h
 - (E) Storm A, but only just at 72 km/h
 - (F) They are exactly equal

100. A figure with order-4 rotational symmetry is turned 90 deg, then a further 180 deg. Does it look the same as at the start, and what total angle did it turn? [\[Symmetry\]](#)

- (A) No, and it turned 360 deg in all
- (B) No, and it turned 90 deg in all
- (C) Yes, and it turned 270 deg in all
- (D) Yes, and it turned 90 deg in all
- (E) Yes, and it turned 180 deg in all
- (F) No, and it turned 270 deg in all